AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 32 · WORKBOOK

Project Problem and Success Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected project plan, with a reproducible-data extension.

State the need, write test inputs, repair vague scope and name non-goals before planning independently.

Keep one habit: Retain versioned requirements, expected cases, decision owner and evidence status.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image and Voice **Pipeline**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Permissions and Human Review

Evidence and Sources

Image and Voice Pipelines

W1 · State purpose and decision owner

> Define the bounded user need and human review.

Your evidence record

- Main planning contract:20-minute limit; name text nonempty/unique; minutes exact int0..60; fun/learning exact int0..5; no Boolean integers; score2fun+learning; highest score then name ascending; reject all occurrences of duplicate names, exclude invalid rows with reasons, distinguish invalid request from no eligible option. Separate pictured chooser uses id,minutes0..30,interest0..5 and scoreinterest,limit8; do not swap these rules.

- State the purpose and decision owner of the worked chooser.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Write concrete tie/order inputs

> Use main-contract records with equal score8.

Your evidence record

- Main planning contract:20-minute limit; name text nonempty/unique; minutes exact int0..60; fun/learning exact int0..5; no Boolean integers; score2fun+learning; highest score then name ascending; reject all occurrences of duplicate names, exclude invalid rows with reasons, distinguish invalid request from no eligible option. Separate pictured chooser uses id,minutes0..30,interest0..5 and scoreinterest,limit8; do not swap these rules.

- Give a tie case for A and B with score 8 each and valid durations.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair vague success

> Replace best with schema, filtering, score and tie behavior.

Your evidence record

- Main planning contract:20-minute limit; name text nonempty/unique; minutes exact int0..60; fun/learning exact int0..5; no Boolean integers; score2fun+learning; highest score then name ascending; reject all occurrences of duplicate names, exclude invalid rows with reasons, distinguish invalid request from no eligible option. Separate pictured chooser uses id,minutes0..30,interest0..5 and scoreinterest,limit8; do not swap these rules.

- Replace the **requirement** always choose the best activity with a checkable one.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · State justified non-goals

> Keep unsupported or unauthorised capabilities outside this prototype.

Your evidence record

- Main planning contract:20-minute limit; name text nonempty/unique; minutes exact int0..60; fun/learning exact int0..5; no Boolean integers; score2fun+learning; highest score then name ascending; reject all occurrences of duplicate names, exclude invalid rows with reasons, distinguish invalid request from no eligible option. Separate pictured chooser uses id,minutes0..30,interest0..5 and scoreinterest,limit8; do not swap these rules.

- Name two capabilities excluded from this **prototype** and explain why.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Plan a new chooser independently

> Keep the first attempt before hints.

Your evidence record

- Use invented records and no live service. Either retain the main schema/score with the requested15-minute limit or state and justify a complete alternative explicitly. Your three acceptance cases must contain actual input records, predicted outputs and reasons, not just names such as boundary or tie. Include the human-review boundary and distinguish a specification from executed evidence.

- Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Plan independently

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Freeze the brief before building so later choices can be traced to **requirement**s.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	

Evidence	My record
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Write an **acceptance case** for Planning the application. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A project recommends from ten approved books using genre and length.

Define **acceptance test**s.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 32 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

My reasoning and evidence

▼ Extension · A reproducible data investigation — W1 State the question

REFERENCE · SECTION 17

Extension · A reproducible data investigation — W1 State the question

> Retained extension task with its own teacher key and response.

Extension question and record

- Write the worked **investigation** question so its population of records is clear.

> **My reasoning and evidence**

▼ **Extension · W2 Trace exclusions**

REFERENCE · SECTION 18

Extension · W2 Trace exclusions

> Use the duration schema beside this standalone task.

Schema, question and response

- Duration extension schema: accept exact integers0..60 including0 and60; reject Boolean,missing,negative or over60 values with reasons. For a mean divide by accepted count only; when none accepted, report no valid observations.

- Classify 0, None, -2 and 60 under the schema.

My reasoning and evidence

▼ **Extension · A reproducible data investigation — W3 Repair a denominator**

REFERENCE · SECTION 19

Extension · A reproducible data investigation — W3 Repair a denominator

> Retained extension task with its own teacher key and response.

Extension question and record

- A report divides a valid total of 24 by six supplied rows, though only three are valid. Correct the valid-record mean.

My reasoning and evidence

▼ Extension · A reproducible data investigation — W4 Investigate independently

REFERENCE · SECTION 20

Extension · A reproducible data investigation — W4 Investigate independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Analyze raw durations [6, 12, None, 0, 70, 18] with the same schema. Report question, accepted list, rejections, mean and one **limitation**.

My reasoning and evidence

▼ Extension · A reproducible data investigation — W5 Support reproduction

REFERENCE · SECTION 21

Extension · A reproducible data investigation — W5 Support reproduction

> Retained extension task with its own teacher key and response.

Extension question and record

- A report gives only mean 9. What additional **evidence** lets a peer reproduce it?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional reading-break chooser needs a transparent contract.

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

AI ACADEMY / AI-2/32

Project Problem and Success Test

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Specify inputs
Validate each row and request; keep rejected-row reasons.



2. Filter and rank
At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.



3. Expected output
A beats tied C by ID; report A and its score 3.



4. Acceptance cases
Also plan invalid request, no eligible item and equal-score cases.

CHANGE ONE THING
Reduce the limit from 8 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Write an equality test and explain which **requirement** it checks.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional reading-break chooser needs a transparent contract.
- Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.
- 1. Filter and rank
- 2. Specify **input**s
- 3. **Acceptance case**s
- 4. Expected **output**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: **Acceptance case**s Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer.

The duration boundary includes equality, and the tie policy prevents accidental list-order dependence. A complete **output** may include why other rows were excluded.

Test expectations should come from the contract rather than copying the first program result.

Changed-case question

- Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Return no eligible item.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: Return no eligible item. All three valid rows take more than five minutes.

Repair and keep evidence

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool **scope**s. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice **Pipeline**s (ai-2/26). Goal: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

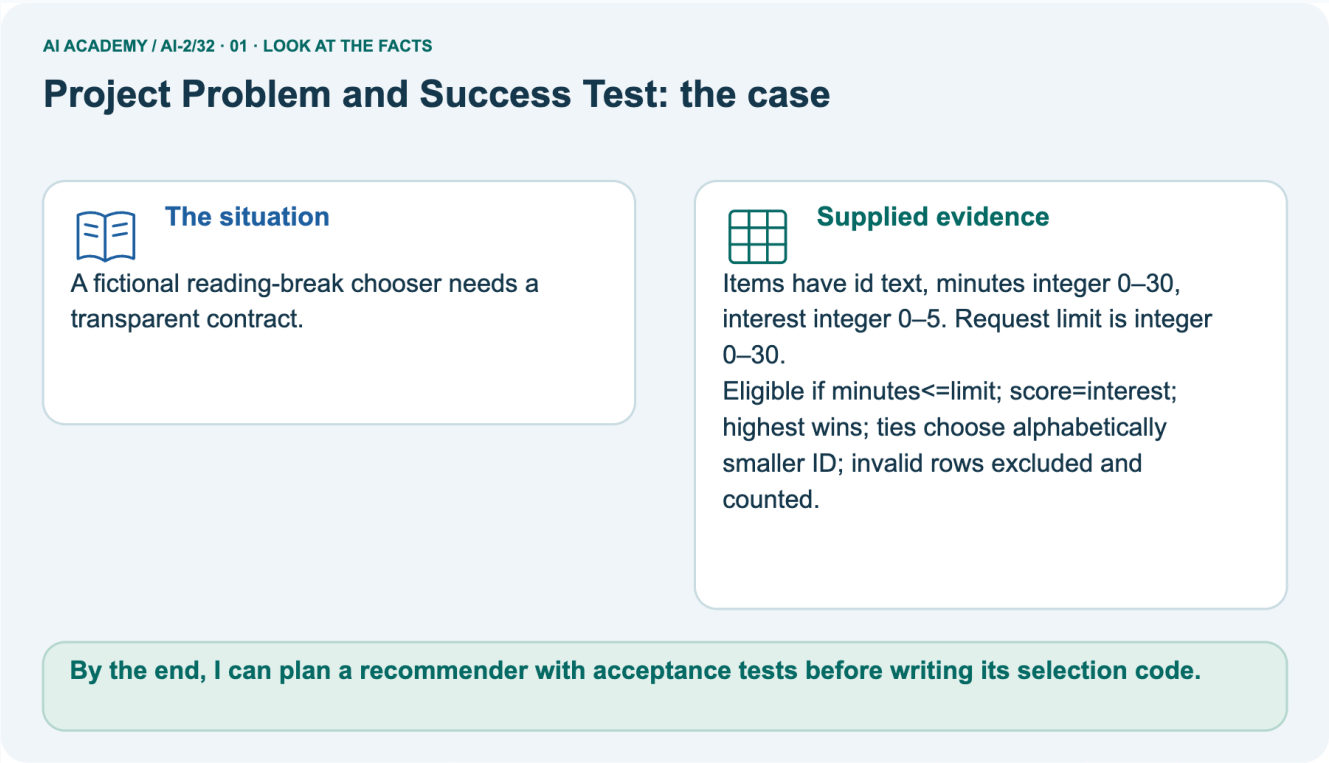
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

Choose the relevant picture rather than repeating every task.

01. Project Problem and Success Test: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

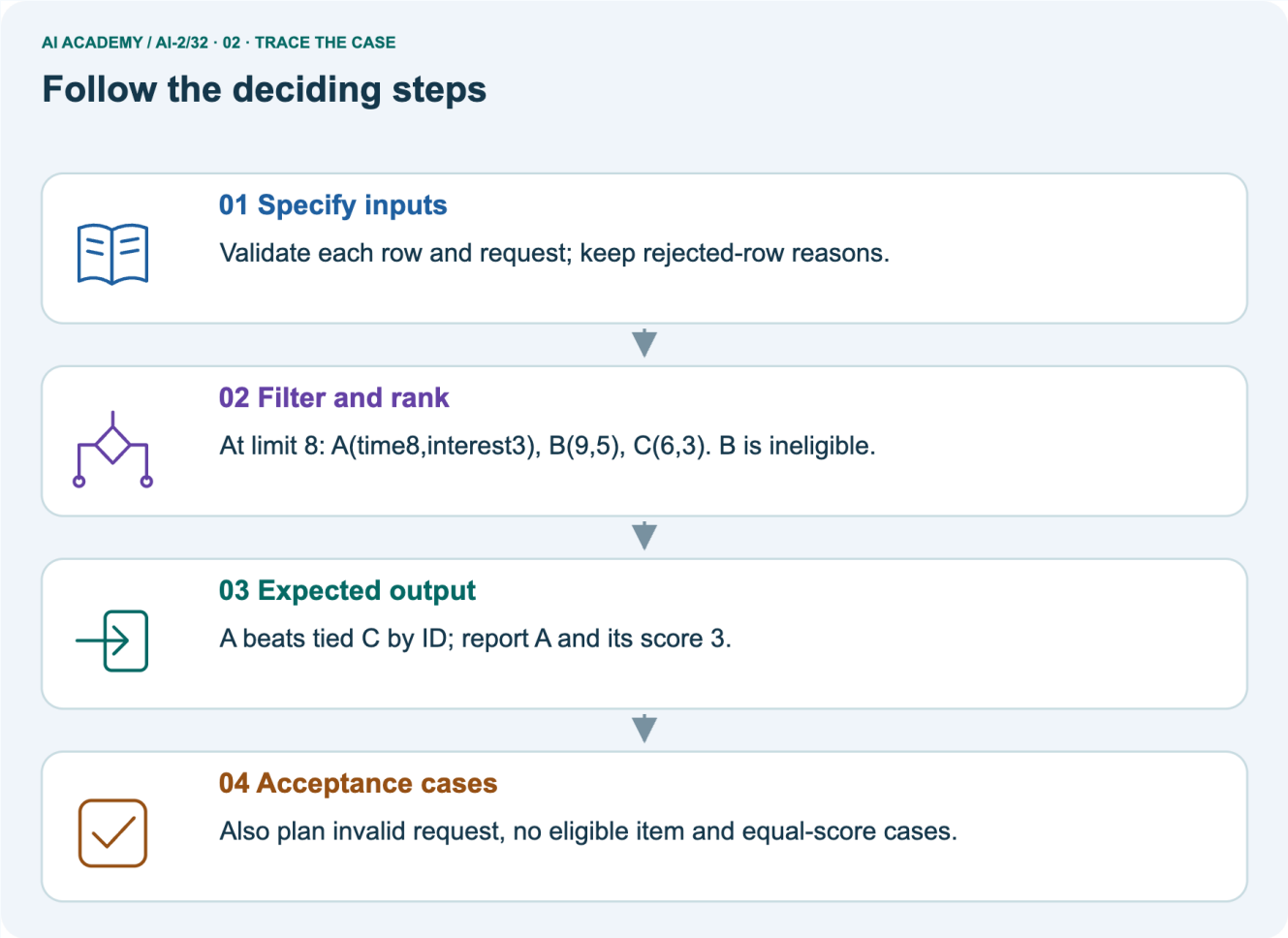
A fictional reading-break chooser needs a transparent contract. Supplied facts: Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

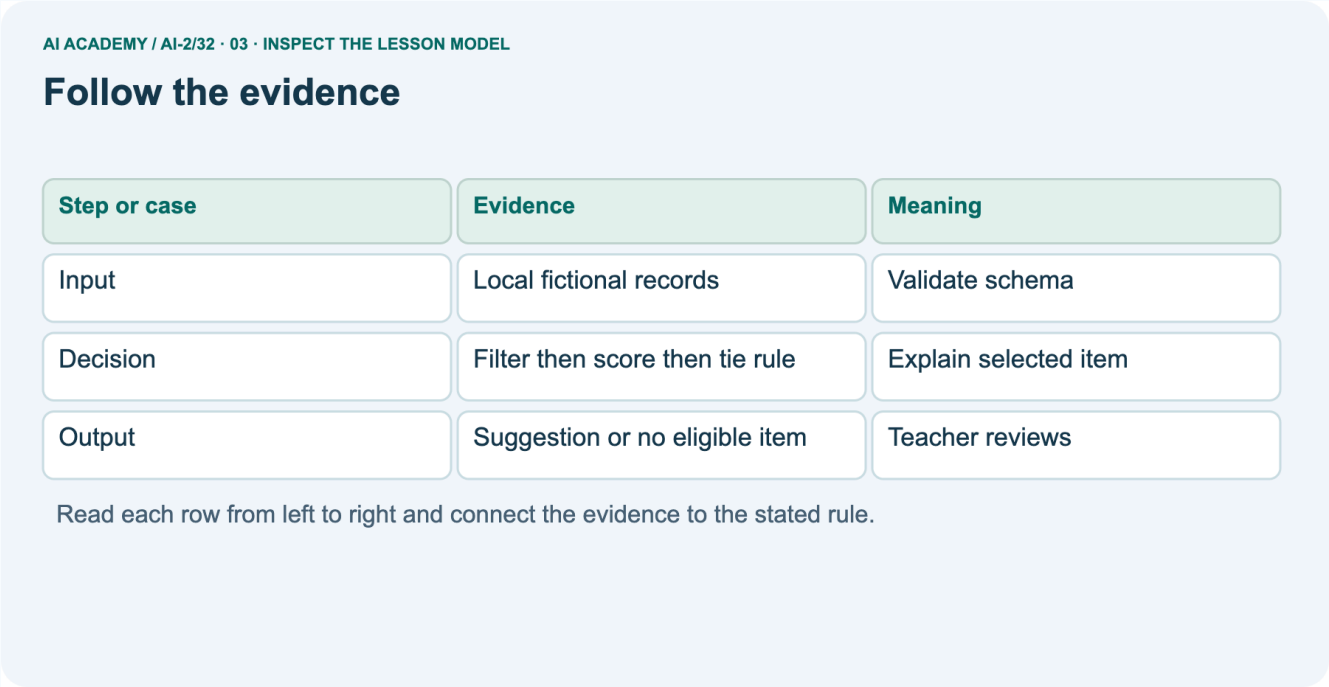
Specify **input**s: Validate each row and request; keep rejected-row reasons. → Filter and rank: At limit 8: A(time8,interest3), B(9,5), C(6,3).

B is ineligible. → Expected **output**: A beats tied C by ID; report A and its score 3. → **Acceptance case**s: Also plan invalid request, no eligible item and equal-score cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



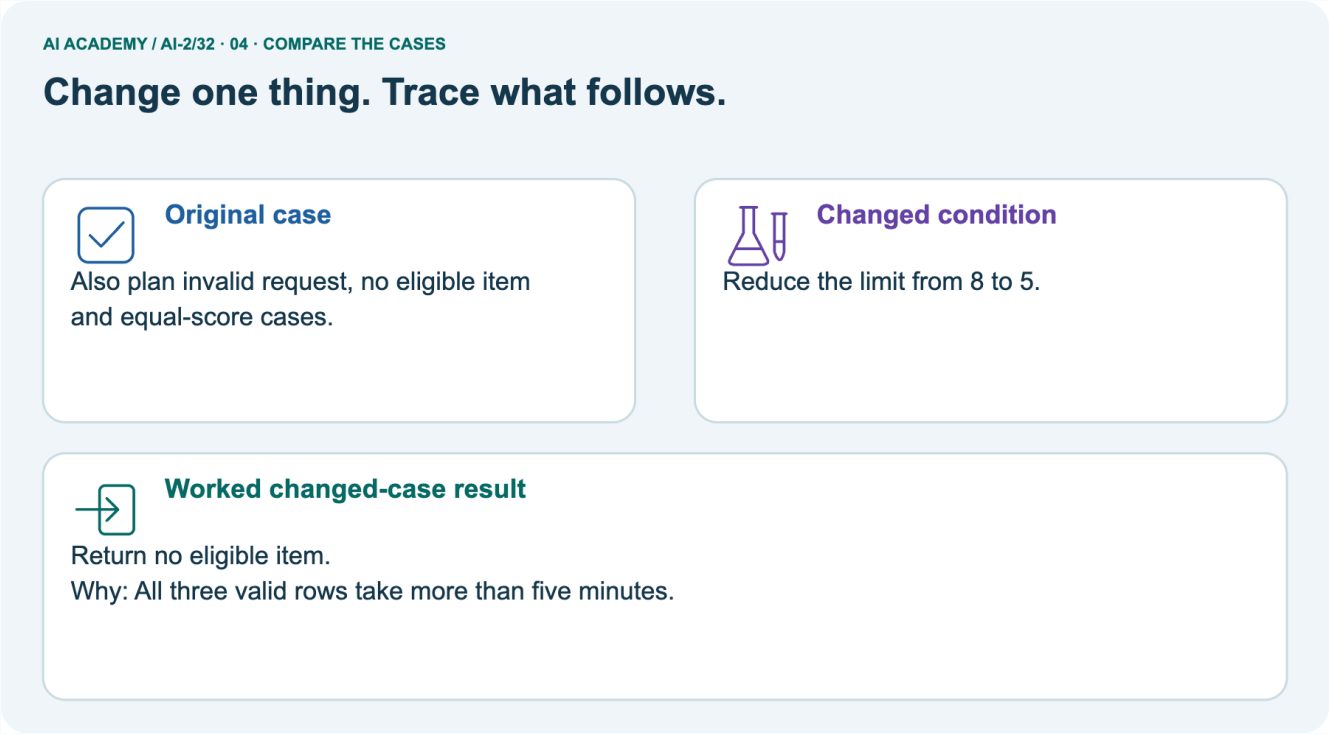
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** | Local fictional records | Validate schema. Decision | Filter then score then tie rule | Explain selected item. **Output** | Suggestion or no eligible item | Teacher reviews

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

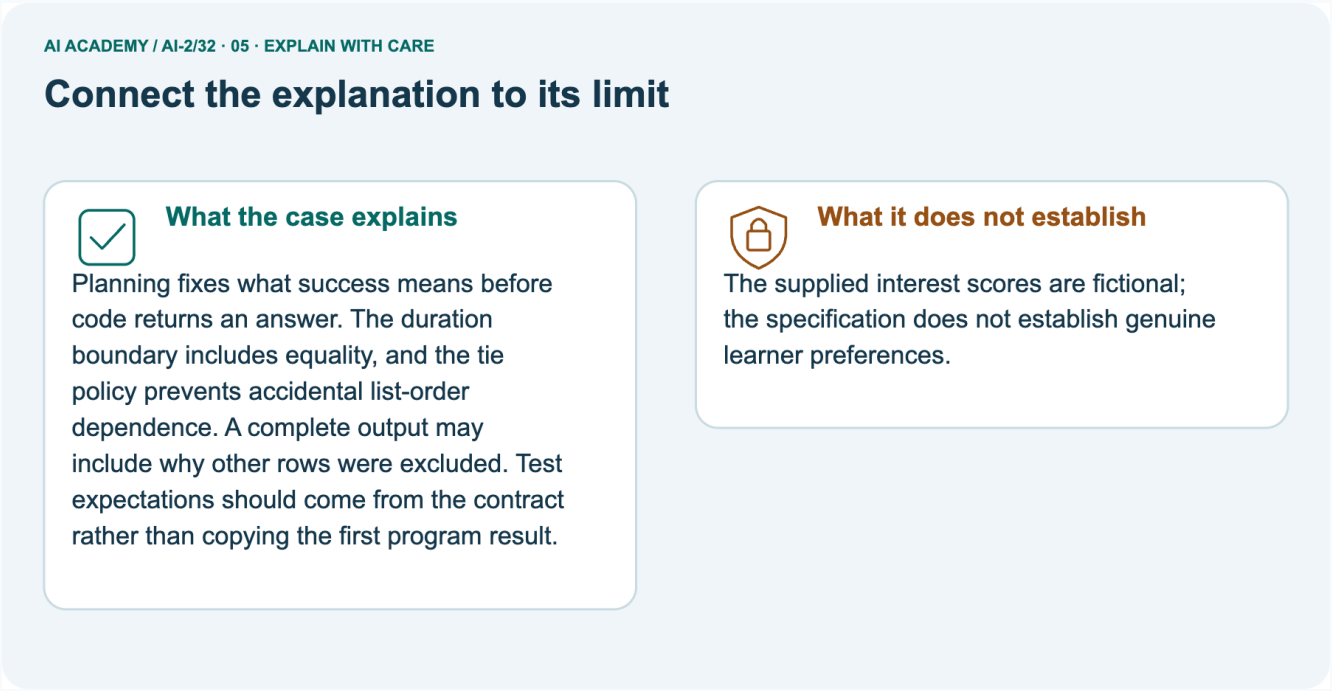
Original case: Also plan invalid request, no eligible item and equal-score cases. Changed condition: Reduce the limit from 8 to 5. Expected result: Return no eligible item.

Reason: All three valid rows take more than five minutes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Planning fixes what success means before code returns an answer. The duration boundary includes equality, and the tie policy prevents accidental list-order dependence.

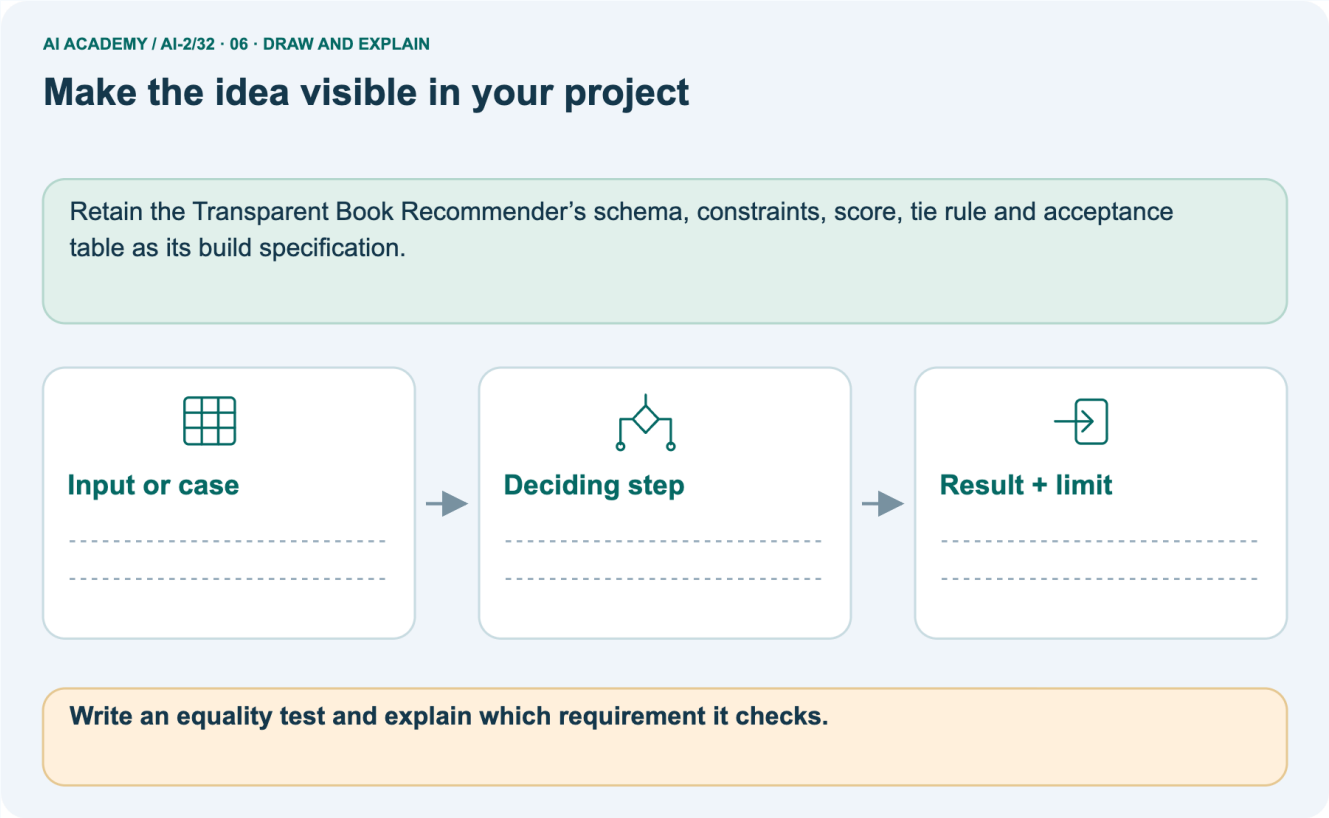
A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Limit: The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification. Draw your **input**, deciding step, result and limit.

Supported question: Write an equality test and explain which **requirement** it checks.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Project Problem and Success Test: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.